

Alex Byun

Sydney, NSW · +(61) 450 370 150 · kimjay2288@gmail.com · github.com/byun-alex · byun-alex.github.io

Final-year Science and Computer Science student building practical AI automation: orchestrating Claude, APIs/MCP connectors, and scripting to turn manual processes into working tools via workflows.

EDUCATION

University of New South Wales - Bachelor of Science and Computer Science

Expected Dec 2026

Relevant coursework: Operating Systems, Computer Networks & Applications, Software Construction, Data Structures & Algorithms, Object-Oriented Design.

TECHNICAL SKILLS

Languages: Python, Bash / shell, HTML, CSS, C, JavaScript / Node.js, Java

AI & Automation: Claude Code (custom skills, hooks, headless orchestration), workflow creation, multi-agent & subagent designs, parallel process management, prompt engineering & versioning, eval harnesses, FFmpeg media pipelines

Agents & skills built: scheduled autonomous cloud agents, a 5-agent parallel roster with group-debate orchestration (AlexOS), 10+ custom Claude Code skills in daily use (session memory, idea ingest, portfolio sync)

MCP & integrations: Zapier MCP (Gmail + Google Calendar automation), Composio MCP (1000+ apps), Telegram Bot API webhooks, Cloudflare Workers + KV, local Stable Diffusion image generation (Fooocus), REST / JSON API wiring & auth

Tools & Platforms: Vercel, Railway (worker + Postgres + Redis), Netlify, Git & GitHub, VS Code, Linux/Unix, Microsoft Office, Excel

Strengths: documentation & process mapping, clear written communication, fast self-directed learning, detail-oriented

PROJECTS

AlexOS - Local Multi-Agent Operating System

github.com/byun-alex/alexos

- Built this as I wanted a team of AI agents that actually work in parallel against my real files. A Node server spawns five Markdown-defined agents (chief-of-staff, research, builder, content, accountability) as real Claude processes; they run concurrently with live status over WebSocket (React, Vite, Tailwind), cost tracked per run.
- A "Mastermind" mode has a panel debate a problem over several rounds, each agent seeing the others' replies, then a chair synthesises the conclusion into shared memory. Security-first: agents can only request actions; the server alone writes memory and disk (*instructions are not capabilities*), so a persona can't be tricked into corrupting state.

Content Factory - Multi-Channel AI Video Production Engine

github.com/byun-alex/content-factory

- Built an engine that turns a vibe and a runtime into a finished multi-hour ambient/lofi video: Claude orchestrates music (**Suno via a Composio MCP connector**), locally generated art (**Stable Diffusion / Fooocus** on my own GPU), and **FFmpeg** assembly, with two human checks so nothing ships unchecked. One engine drives multiple channels through swappable style guides; generation is tiered by budget (local GPU / cheap API / manual fallback) so a missing connector pauses the run instead of killing it.
- Cut the visual step's cost to zero by replacing a paid video API with a synthesized rain overlay composited over local stills.

Morning Brief Agent - Autonomous, Spec-Driven Daily Agent

github.com/byun-alex/morning-brief-agent

- A scheduled cloud agent that every morning reads my calendar, inbox, deadlines and weather, composes a manager-style brief, and emails it. It replaced a repetitive manual task I did by hand, and it runs into my own inbox on a cron schedule with no app to open.
- The whole behaviour is a Markdown spec read at run time, so adding a section is an edit, not a redeploy. Wired to Gmail and Calendar through an MCP connector; if a source is empty or a tool fails, the brief says so plainly instead of inventing.

Instagram Comment-to-DM Automation - Self-Hosted, Live in Production

[my fork](#)

- Replaced a \$14/month SaaS billed per person contacted with a self-hosted deployment at about \$5/month, uncapped. Deployed an open-source clone (**diwenne/openreply**, not my application code) end to end: **Next.js on Vercel** for dashboard, OAuth and Meta webhooks, plus a **queue worker with Postgres and Redis on Railway**, split out because a queue consumer must stay awake and serverless sleeps.
- Root-caused a live incident on a post at 57,000 views (duplicate DMs, every send logging FAILED): the cause was not my code but a second tool still connected to the same account, and the tell was a missing emoji on the duplicate's button. Fixed two real bugs on the way, a retry storm and a webhook/poll race, making the send-log insert the atomic claim.

More on GitHub: **log-triage** (support triage CLI, 17 tests) · **murmur** (on-device dictation) · **market-data-scanner** (async market-data engine, 49 tests) · **overnight-agent** (verify-gated job runner) · **telegram-idea-capture** (Cloudflare Worker + KV inbox) · **brain-dashboard** (Node 24 PTY root-cause).

EXPERIENCE

AI Trainer - Outlier (Freelance / Contract)

May 2024 – Mar 2026

- Review and evaluate AI-generated responses (text, code, images, design) for accuracy, clarity, and relevance; document errors with consistent, detail-focused written feedback to improve model output.

Customer Service, Social Media & Admin - Woolworths · Oliver Brown · a Shopify business

2021 – 2023

- Handled frontline customer support, bookings, and accurate record-keeping in fast-paced environments, staying calm and methodical under pressure; managed TikTok, Instagram, and YouTube for a small business, using Shopify analytics to adjust ad planning.

ADDITIONAL

Languages: English (native), Korean (proficient), Japanese (intermediate) **Eligibility:** Australian citizen · Full driver's licence